

Peter Ramsey

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Curriculum Vitae

Research Interests

Topics: Algebraic combinatorics, algebraic topology, & computational algebra.

Objects: Matroids, hyperplane arrangements, & generalizations.

Education

PhD Mathematics

(Expected) May 2027

Louisiana State University – Baton Rouge, LA

Advisor – Dr. Christin Bibby

MS Mathematics

May 2023

Louisiana State University – Baton Rouge, LA

BA Pure Mathematics & Sociology

May 2021

Spring Hill College – Mobile, AL

Publications

Accepted

The Orlik–Solomon algebra of a locally geometric poset

with Christin Bibby.

Peer reviewed extended abstract; to appear in Séminaire Lotharingien de Combinatoire, Proceedings of FPSAC 2026.

Submitted for publication

The Orlik–Solomon algebra of a locally geometric poset and cohomology of complex abelian arrangements

with Christin Bibby.

Conference Participation

Invited Talk: Title to be announced

October 2026

Special Session on Algebraic and Geometric Aspects of Matroid Theory, AMS Fall Central Sectional Meeting, University of Minnesota-Duluth, Duluth, MN.

Invited Talk: The Orlik–Solomon algebra of a matroid prescheme

September 2026

LSU Combinatorics Seminar, Louisiana State University, Baton Rouge, LA.

Poster Presentation: The Orlik–Solomon algebra of a locally geometric poset

July 2026

38th International Conference on Formal Power Series and Algebraic Combinatorics, University of Washington, Seattle, WA.

Participant: Topology of arrangements with an eye to applications

September 2025

International Research Conference, University of Pisa, Pisa, Italy.

Participant: Séminaire de Mathématiques Supérieures (SMS) 2025: An Introduction to Recent Trends in Commutative Algebra

June 2025

SLMath Summer School, Fields Institute, Toronto, Canada.

Honors & Awards

Research Training Group (RTG) Graduate Fellow

May 2024 – August 2026

Graduate fellowship awarded to dedicate time towards research.

Certificate of Teaching Excellence

May 2024, May 2023

Awarded to select graduate teaching assistants in the mathematics department for outstanding teaching ability.

Teaching Experience

Instructor of Record: Differential and Integral Calculus

Fall 2026

Responsible for developing lesson plans, preparing assignments, delivering class lectures, and assigning final grades.

Instructor of Record: College Algebra

Fall 2022 – Spring 2024

Responsible for delivering class lectures, tutoring in the LSU Math Lab, and assigning final grades.

Grader

Fall 2021 – Spring 2024

Graded for courses in linear algebra, multidimensional calculus, differential equations, probability, and applied algebra.

Professional Service

Graduate Mentor: Summer Math Workshop

May 2025

Supervised and mentored a small group of undergraduate students, facilitating discussion as they learned about selected math topics.

Graduate Mentor: Directed Reading Program

Fall 2023

Mentored an undergraduate student in chess board structures in graph theory.

Co-Chair: Geaux Mathematics Committee

Summer 2023

Organized and moderated sessions for teaching, research, and mental health during incoming graduate student orientation.

Graduate Teaching Mentor: Communicating Mathematics

Spring 2023 – Spring 2024

Supervised and mentored new graduate students on instructing college algebra in a classroom setting.

Facilitator: ALEKS Test Preparatory Camp

Summer 2022

Responsible for preparing and delivering lectures and facilitating group work sessions for undergrads preparing for the ALEKS placement test in mathematics.